In [1]:
```
print("Hello world...")
```

```
Hello world...
```

In [3]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

In [11]:
```
train = pd.read_csv("C:/Users/Asus/Downloads/titanic/train.csv")
test = pd.read_csv("C:/Users/Asus/Downloads/titanic/test.csv")
```

In [ ]:

In [14]:
```
train.head()
```

Out[14]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

In [16]:
```
test.head()
```

Out[16]:

	PassengerId	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	892	3	Kelly, Mr. James	male	34.5	0	0	330911	7.8292	NaN	Q
1	893	3	Wilkes, Mrs. James (Ellen Needs)	female	47.0	1	0	363272	7.0000	NaN	S
2	894	2	Myles, Mr. Thomas Francis	male	62.0	0	0	240276	9.6875	NaN	Q
3	895	3	Wirz, Mr. Albert	male	27.0	0	0	315154	8.6625	NaN	S
4	896	3	Hirvonen, Mrs. Alexander (Helga E Lindqvist)	female	22.0	1	1	3101298	12.2875	NaN	S

In [18]:

```
train.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

In [20]:

```
train.describe()
```

Out[20]:

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

In [22]:

```
# Missing Values
train.isnull().sum()
```

Out[22]:

```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

In [24]:

```
test.isnull().sum()
```

Out[24]:
```
PassengerId      0
Pclass           0
Name             0
Sex              0
Age             86
SibSp            0
Parch            0
Ticket           0
Fare             1
Cabin          327
Embarked         0
dtype: int64
```

In [ ]:

```

```

In [28]:
```
# Summary statistics
train.describe()

# Info about data types and missing values
train.info()

# Value counts of a categorical column (example: 'Sex')
train['Sex'].value_counts()
```

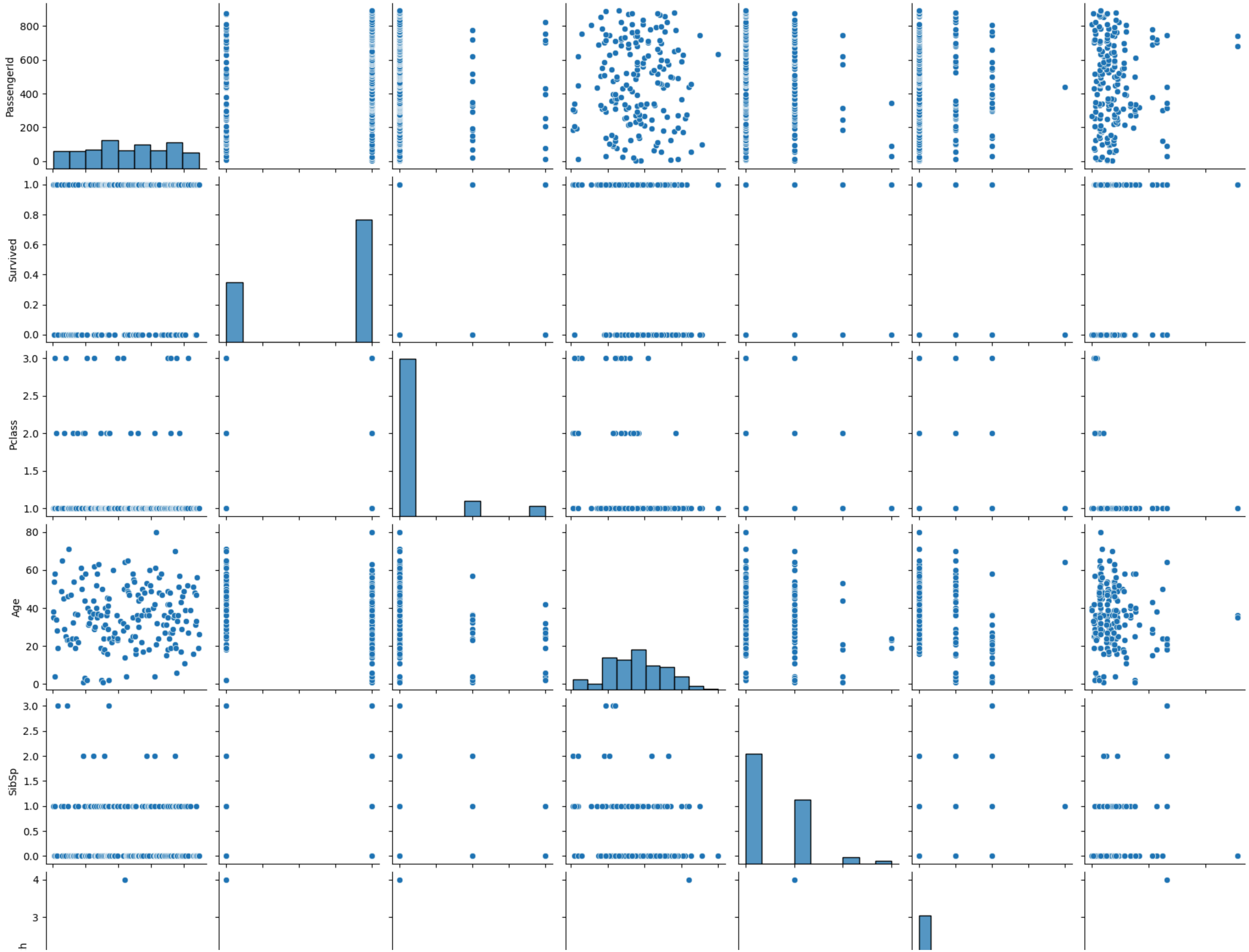
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   PassengerId  891 non-null    int64  
 1   Survived     891 non-null    int64  
 2   Pclass       891 non-null    int64  
 3   Name         891 non-null    object 
 4   Sex          891 non-null    object 
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64  
 7   Parch        891 non-null    int64  
 8   Ticket       891 non-null    object 
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object 
 11  Embarked     889 non-null    object 
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

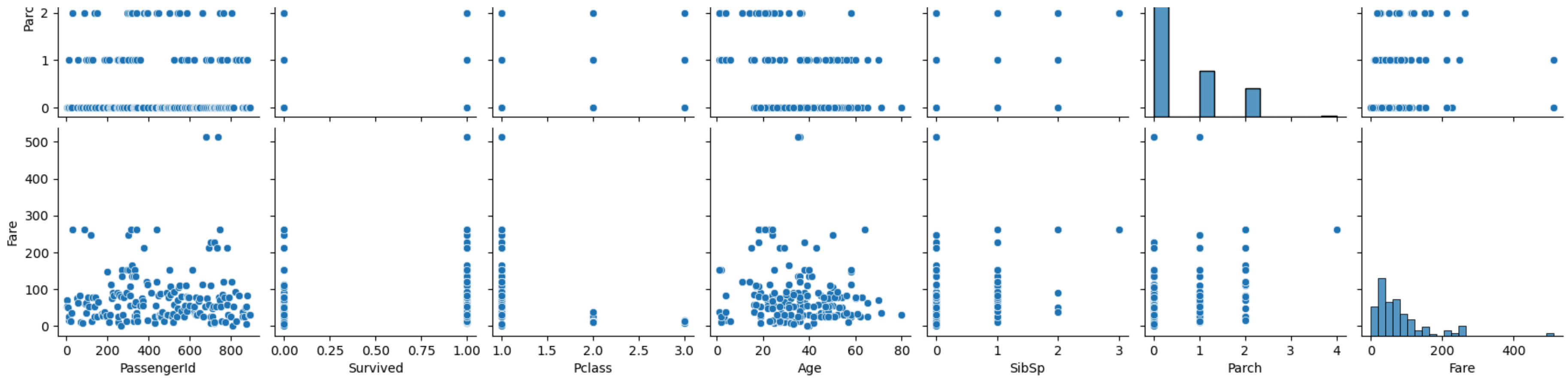
Out[28]:

```
Sex
male      577
female    314
Name: count, dtype: int64
```

In [30]:

```
# Pairplot (best for smaller datasets)
sns.pairplot(train.dropna())  # dropna to avoid errors due to NaNs
plt.show()

# Heatmap of correlation
sns.heatmap(train.corr(), annot=True, cmap='coolwarm')
plt.title('Correlation Heatmap')
plt.show()
```


PassengerId
Survived
Pclass
Age
SibSp



Parc
Fare
PassengerId
Survived
Pclass
Age
SibSp
Parch
Fare

```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
Cell In[30], line 6
      3 plt.show()
      5 # Heatmap of correlation
----> 6 sns.heatmap(train.corr(), annot=True, cmap='coolwarm')
      7 plt.title('Correlation Heatmap')
      8 plt.show()

File C:\ProgramData\anaconda3\Lib\site-packages\pandas\core\frame.py:11049, in DataFrame.corr(self, method, min_periods, numeri
c_only)
  11047 cols = data.columns
  11048 idx = cols.copy()
> 11049 mat = data.to_numpy(dtype=float, na_value=np.nan, copy=False)
  11051 if method == "pearson":
  11052     correl = libalgos.nancorr(mat, minp=min_periods)

File C:\ProgramData\anaconda3\Lib\site-packages\pandas\core\frame.py:1993, in DataFrame.to_numpy(self, dtype, copy, na_value)
   1991 if dtype is not None:
   1992     dtype = np.dtype(dtype)
-> 1993 result = self._mgr.as_array(dtype=dtype, copy=copy, na_value=na_value)
   1994 if result.dtype is not dtype:
   1995     result = np.asarray(result, dtype=dtype)

File C:\ProgramData\anaconda3\Lib\site-packages\pandas\core\internals\managers.py:1694, in BlockManager.as_array(self, dtype, c
opy, na_value)
   1692         arr.flags.writeable = False
   1693 else:
-> 1694     arr = self._interleave(dtype=dtype, na_value=na_value)
   1695     # The underlying data was copied within _interleave, so no need
   1696     # to further copy if copy=True or setting na_value
   1698 if na_value is lib.no_default:

File C:\ProgramData\anaconda3\Lib\site-packages\pandas\core\internals\managers.py:1753, in BlockManager._interleave(self, dtyp
e, na_value)
   1751     else:
   1752         arr = blk.get_values(dtype)
-> 1753     result[rl.indexer] = arr
   1754     itemmask[rl.indexer] = 1
   1756 if not itemmask.all():
```

```
ValueError: could not convert string to float: 'Braund, Mr. Owen Harris'
```

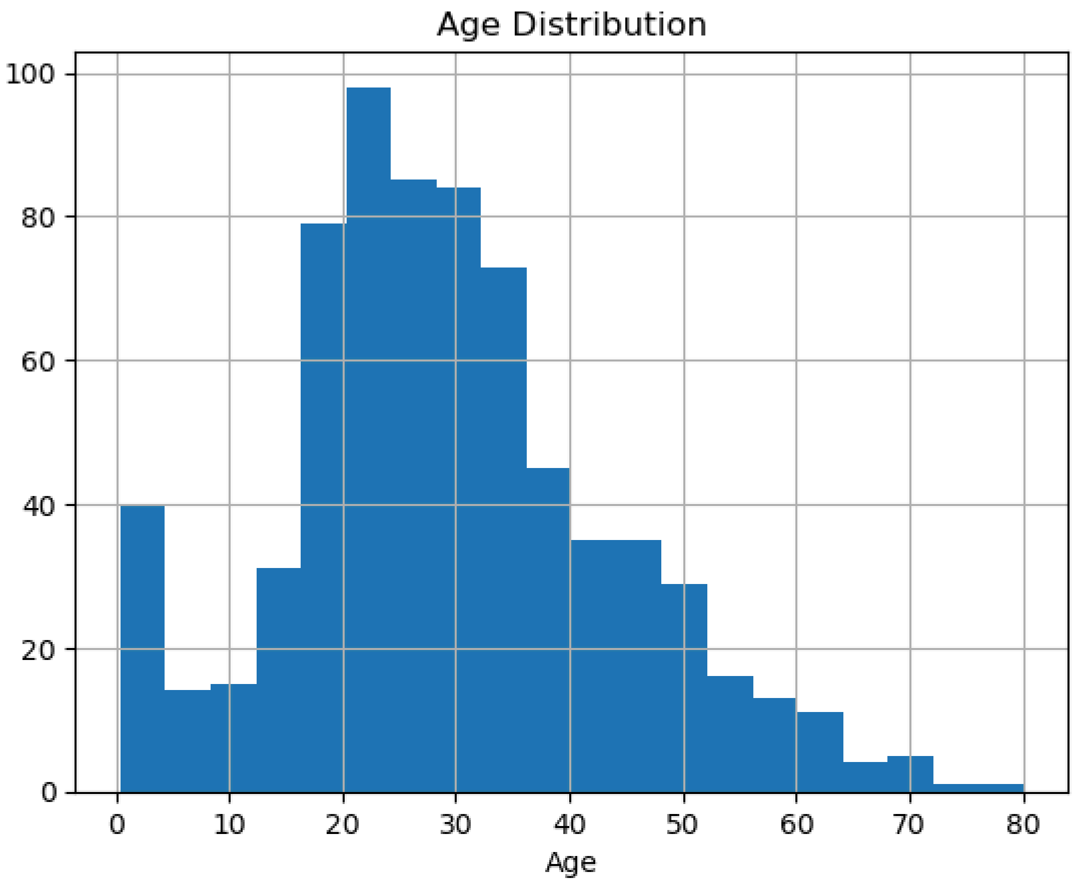
In [ ]:

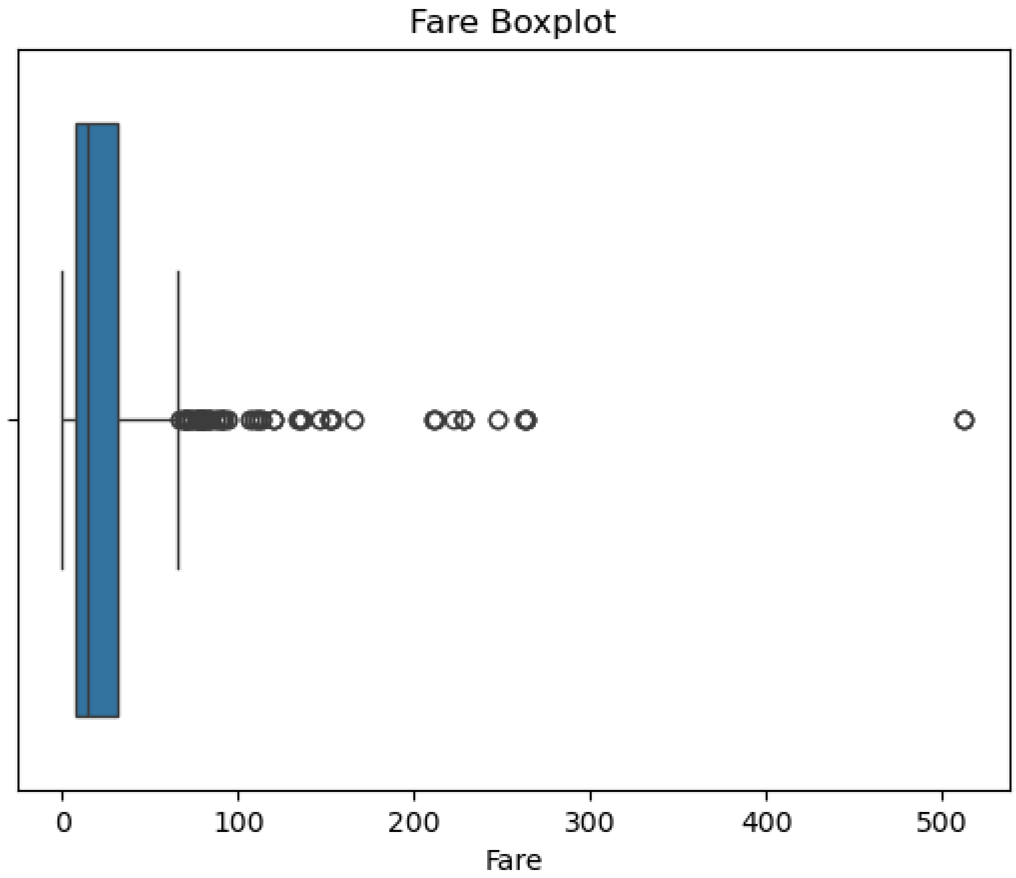
In [32]:
```python
# Survival rate by gender
print(train.groupby('Sex')['Survived'].mean())

# Survival rate by passenger class
print(train.groupby('Pclass')['Survived'].mean())

# Average age by survival
print(train.groupby('Survived')['Age'].mean())
```

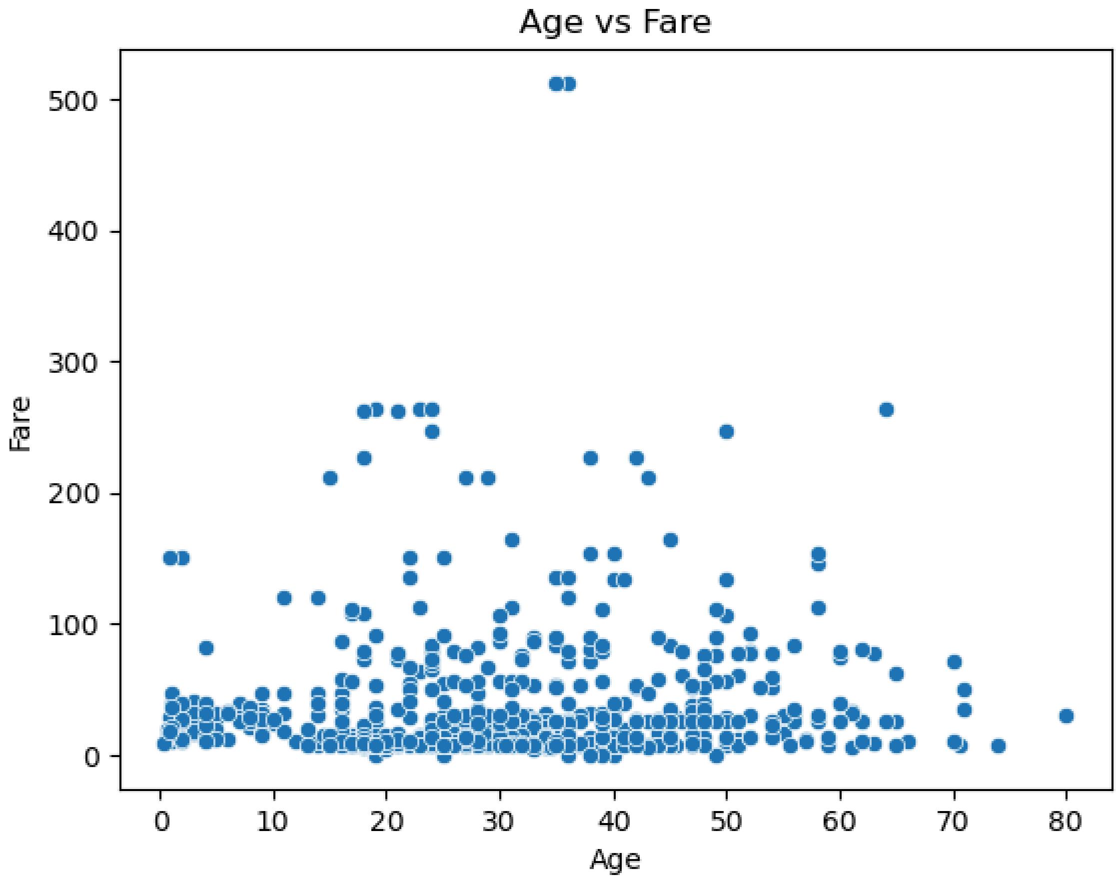
```
Sex
female    0.742038
male      0.188908
Name: Survived, dtype: float64
Pclass
1    0.629630
2    0.472826
3    0.242363
Name: Survived, dtype: float64
Survived
0    30.626179
1    28.343690
Name: Age, dtype: float64
```

In [34]:
```python
# Histogram of Age
train['Age'].hist(bins=20)
plt.xlabel('Age')
plt.title('Age Distribution')
plt.show()

# Boxplot of Fare
sns.boxplot(x=train['Fare'])
plt.title('Fare Boxplot')
plt.show()

# Scatterplot: Age vs Fare
sns.scatterplot(x='Age', y='Fare', data=train)
```

```python
plt.title('Age vs Fare')
plt.show()
```



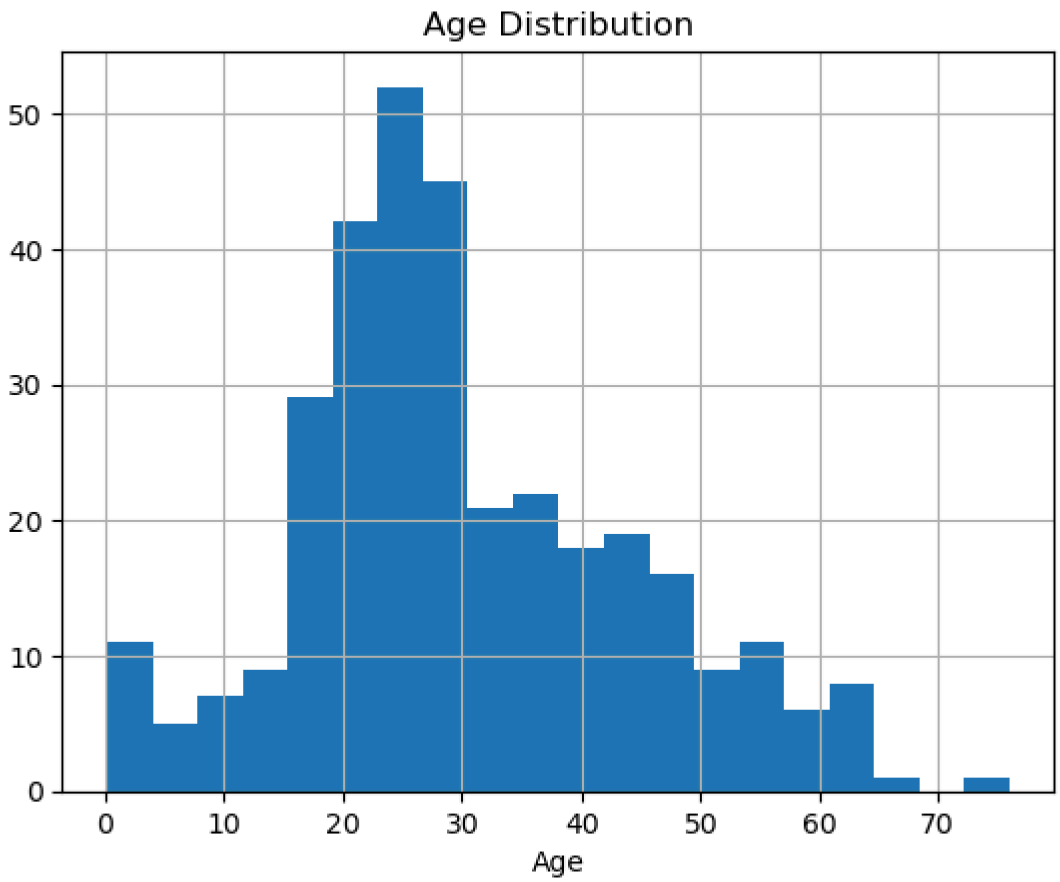

Fare Boxplot
0
100
200
300
400
500
Fare

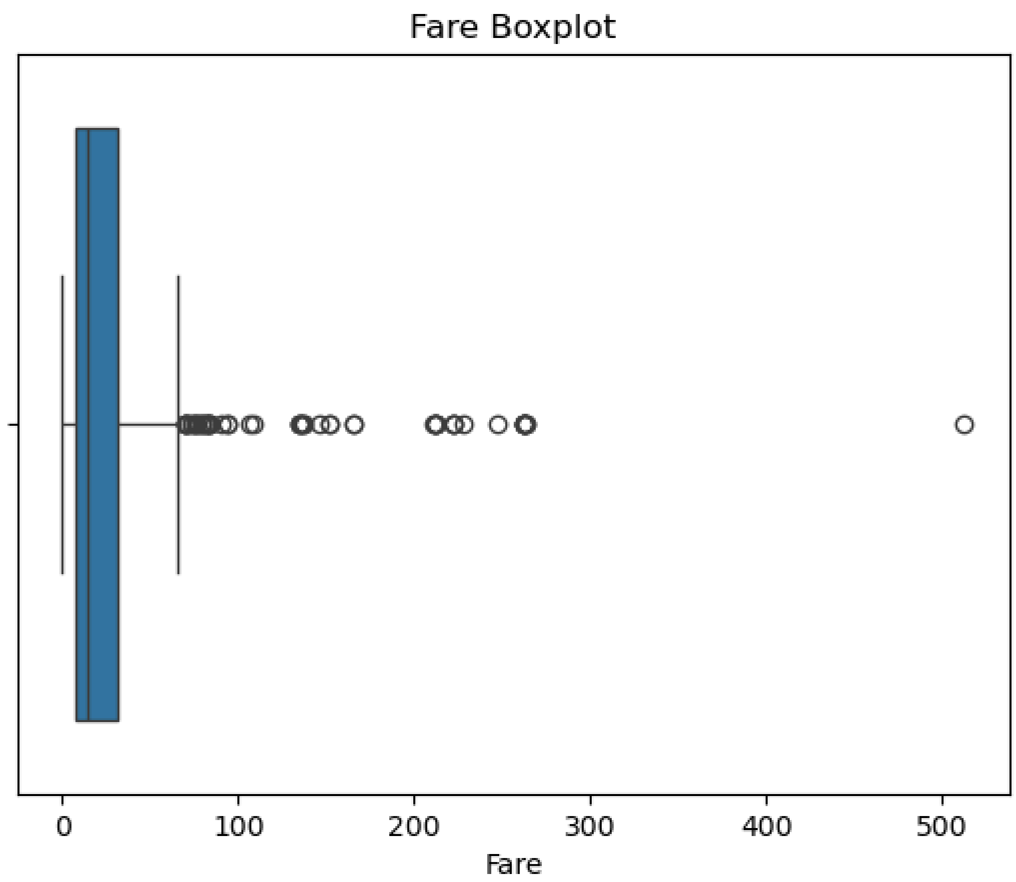


In [36]:

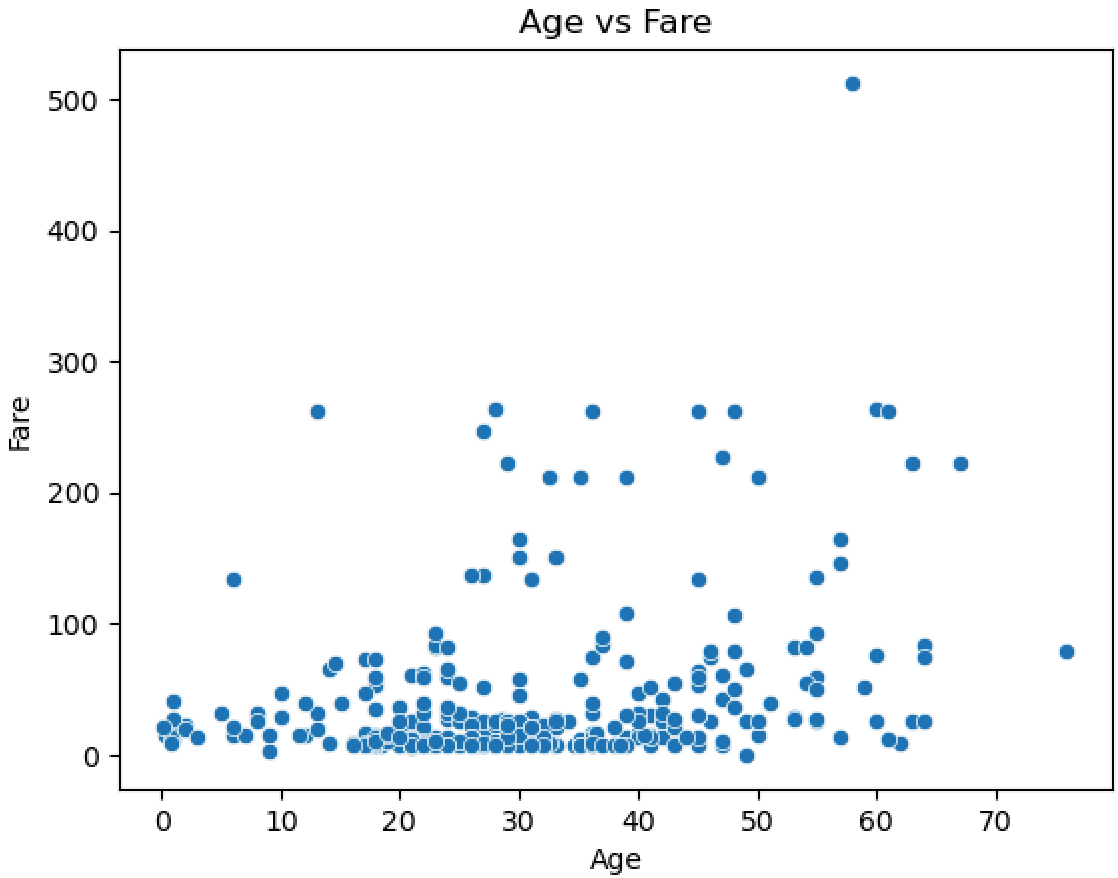
```
# Histogram of Age
test['Age'].hist(bins=20)
plt.xlabel('Age')
plt.title('Age Distribution')
plt.show()

# Boxplot of Fare
sns.boxplot(x=test['Fare'])
plt.title('Fare Boxplot')
plt.show()

# Scatterplot: Age vs Fare
sns.scatterplot(x='Age', y='Fare', data=test)
```

```
plt.title('Age vs Fare')
plt.show()
```





Fare Boxplot
0
100
200
300
400
500
Fare



In [ ]: